

João Rietra

Researcher and Lecturer · AI model reliability: calibration, out-of-distribution generalisation and explanation faithfulness

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SUMMARY

Researcher in AI model reliability, focused on calibration, out-of-distribution generalisation and explanation faithfulness. I compare models under a controlled protocol, multiple seeds and statistical tests (confidence intervals, Wilcoxon), and I report what the method supports. I author five works between dissertation and papers, three submitted to SBTI 2026, from medical imaging to manipulation detection and tree-ensemble adaptation. I teach programming, statistics and data visualisation as a teaching assistant at CESAR School.

KEYWORDS

Model reliability · model calibration (ECE) · out-of-distribution evaluation (OOD) · explanation faithfulness (Grad-CAM, occlusion) · experimental design · reproducibility · inferential statistics (CI, Wilcoxon) · transfer learning · medical imaging (MedMNIST) · computer vision · image-manipulation detection · tree-ensemble adaptation · foundation models · Python · PyTorch · MLflow · technical mentoring and teaching · domain-expert validation

EDUCATION

- **Professional Master's in Software Engineering** — CESAR School, Recife · Oct 2025 – defence expected Nov 2026
 - **Technologist Degree in Data Science** (3-year undergraduate) — CESAR School, Recife · Jan 2025 – expected completion Jul 2027
 - **Doctor of Medicine (MD)** — FPS / IMIP, Recife · completed 2022 · CRM-PE 32980
 - **Science Without Borders exchange (Neuroscience)** — University of Alberta, Canada · Aug 2015 – Jul 2016
 - **Computer Science** — CIn / UFPE · Jan 2014 – interrupted
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EXPERIENCE

AI Engineer — LetzGrow (School Innovation, CESAR School) · 2026 – present

AI career platform with versioned fairness evaluation of its recommendation pipeline. Selected for 2026.2, as a two-person team.

Technical Lead (data and computer vision) — MAPEN (School Innovation, CESAR School) · 2026 – present

Nationwide energy-loss detection, selected for 2026.2.

- **Nationwide data base** (hourly grid-operator series 2000→2026, billed consumption by state and IBGE boundaries in DuckDB), with documented provenance per dataset (source, author, licence, granularity).
- **Automatic meter reading**: fine-tuned TrOCR-small on the UFPR-AMR dataset, benchmarked against PP-OCRv6-tiny, with MLflow tracking.

ML Engineer — Veritas (School Innovation, CESAR School) · 2025 – 2026

- **Applied research on deepfake detection**: Grad-CAM heatmap, systematic false-positive analysis and out-of-distribution (OOD) evaluation.

Software Engineer (Clinical Domain + QA) — GasoGraph (Master's / CESAR) · 2025 – 2026

- **Rules from published consensus:** turned veterinary arterial blood-gas consensus into clinical decision flowcharts and implemented the rule engine, with expert-domain validation and clinically driven QA.

Teaching Assistant — CESAR School · 2025 – 2026

- Programming (C), Statistics (R), Calculus I, Competitive Programming and Data Analysis and Visualization.

Medical Data Analyst — J & L Hospital Consulting · 2023 – 2024

- **Clinical outcomes and quality:** measured readmission, hospital-acquired infection and post-operative complications, flagged failures for protocol adjustment and tracked ONA and JCI accreditation KPIs.

Physician — Pediatric Emergency and Primary Care · 2022 – 2025

- Diagnostic decisions under high demand in a paediatric emergency department and longitudinal care in primary care.

Researcher — Computational Neuroscience, University of Alberta · 2015 – 2016

- Science Without Borders exchange: neural-flow modelling, high-dimensional datasets and applied bioinformatics (indel identification in BAM files, `indelMINER`).

ACADEMIC OUTPUT

1. *Training Reuse for Binary Triage in Multi-Domain Medical Imaging* (Master's dissertation, CESAR School, 2026): controlled comparison of 4 backbone origins across 5 heterogeneous domains (MedMNIST), evaluating AUC, calibration (ECE) and occlusion-based explanation faithfulness. Forensic intermediate fine-tuning won (AUC 0.842 vs. 0.796 for ImageNet) with far better calibration (ECE 0.045 vs. 0.219). Own audit protocol for explanation faithfulness; MLflow and statistics (CI, Wilcoxon) over multiple seeds.
2. *Asymmetric generalisation in image-manipulation detectors: from the training domain to collapse on manual splicing* (SBTI 2026, submitted): empirical and adversarial evaluation of own (EfficientNet-B0) and external (ViT, Swin) detectors on 3 distributions.
3. *Fine-tuning decision forests under drift: a Bayesian blend with a unified vocabulary* (SBTI 2026, submitted): adaptation of Random Forest, XGBoost and LightGBM to new data without retraining a single tree, validated on 4 public benchmarks (Folktables/ACS, Electricity, Covertypes, rotated MNIST).
4. *EfficientNeXt: an autoregressive mini-LLM derived from the EfficientNet architecture with unified generation and embedding (GRIT) and decoding steering without an auxiliary model.*
5. *Automated inspection of electricity-meter reading photos: an OCR pipeline and characterisation of the coherence-validation problem* (SBTI 2026, submitted).

Papers written in Portuguese; original titles are in the Portuguese version of this CV.

PROJECTS

btc-forecast — foundation models and fine-tuning · github.com/jhlr/BTC-Forecast

Systematic comparison of 5 zero-shot time-series foundation models (Chronos, TimesFM, Moirai, Lag-Llama, TTM) in a reproducible fine-tuning, ensembling and directional-backtesting pipeline, with MLflow tracking.

Medical school thesis: diabetes in patients undergoing amputation

Retrospective observational study of 104 patients (IMIP, 2011–2015): chart collection, statistical analysis (stepwise regression) and comparison with national and international literature.

SKILLS

Research and machine learning: experimental design · reproducibility · reliability evaluation (calibration, OOD, explanation faithfulness) · inferential statistics (CI, Wilcoxon) · critical reading of literature and consensus · scientific writing · PyTorch · transfer learning and fine-tuning · medical imaging · computer vision · time series and foundation models · MLflow · Python · SQL

Languages: Portuguese (native) · English (native) · Spanish (basic) · French (basic)